Bryan Kian Sarmen

Angat, Bulacan, Philippines | +63 994 024 9061 | sarmenbryankian@gmail.com

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 3+ years of professional experience designing, building, and maintaining production web applications and REST APIs. Skilled in TypeScript, React, Next.js, Node.js, Java, Kotlin, FastAPI, PostgreSQL, Supabase, Docker, and AWS/Kubernetes cloud environments. Experienced in secure multi-tenant architecture, Stripe payment integrations, asynchronous processing, retrieval-augmented generation (RAG) systems, automated testing, and CI/CD pipeline maintenance. Proven ability to independently design and ship complete production-ready applications from database schema through deployment and monitoring.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, Kotlin, C#

Frontend: React, Next.js, React Router, Tailwind CSS, Vite

Backend: Node.js, Express, FastAPI, Spring Boot, Ktor, ASP.NET Core, Apache Camel

Databases: PostgreSQL, Supabase, MongoDB, Redis, MySQL, Microsoft SQL Server

APIs & Architecture: REST APIs, server-side rendering (SSR), layered architecture, microservices, asynchronous processing, event-driven workflows (Redis Streams)

Authentication & Authorization: Supabase Auth, Supabase Row Level Security (RLS), OAuth-based workflows

Payments & Integrations: Stripe (webhooks, checkout), Resend, OpenRouter

AI & Search: Retrieval-augmented generation (RAG), Chroma Cloud, hybrid search, dense/sparse vectors, Reciprocal Rank Fusion

Testing: Unit testing, integration testing, component testing, end-to-end testing, Playwright, JUnit, Mockito

Cloud & DevOps: Docker, Kubernetes, AWS, Render, Vercel, Jenkins, GitLab CI/CD, Maven

Monitoring & Observability: Grafana, Prometheus, Sentry, Elasticsearch

Tools: Git, GitHub, Jira, GitHub Copilot

PROFESSIONAL EXPERIENCE

Software Engineer — Digital Services Cambridge Limited ROHQ

Makati, Metro Manila, Philippines | January 2025 – Present

- Develop and maintain REST APIs and microservices using Java, Spring Boot, Apache Camel, and Node.js.
- Maintain CI/CD pipelines and cloud infrastructure using AWS, Kubernetes, Docker, Jenkins, GitLab CI/CD, and Maven, supporting automated build and deployment workflows.
- Optimize Elasticsearch indexing strategies to improve application stability and search performance.
- Build Python automation scripts and Playwright end-to-end tests to support engineering and quality-assurance workflows.
- Monitor and troubleshoot production systems using Grafana, Prometheus, and Sentry to identify and resolve issues.

Software Engineer — Marlow Navigation Philippines Inc.

Malate, Metro Manila, Philippines | April 2023 – January 2025

- Developed REST APIs using Kotlin, Ktor, C#, and ASP.NET Core to support core backend service functionality.
- Improved backend latency through database query optimization and connection pooling.
- Diagnosed and resolved backend stability issues related to object garbage collection and out-of-memory conditions, improving service reliability.
- Built automation scripts and contributed to Windows-based kiosk applications.

PROJECTS

Quick Reserve — Full-Stack Marketplace (Independent Project)

Technologies: Next.js 16, App Router, TypeScript, React 19, Tailwind CSS v4, Supabase, PostgreSQL, Row Level Security, Supabase Auth, Supabase Storage, Stripe, Node.js, Express, Resend

Links: github.com/bryankiansarmen/quick-reserve-web | github.com/bryankiansarmen/quick-reserve-webhook-service | quick-reserve-psi.vercel.app

- Built a two-sided marketplace enabling space owners to list bookable venues and renters to search, book, and pay for time slots, using Next.js 16 App Router and Supabase/PostgreSQL.
- Implemented an atomic booking flow with database-level availability locking to prevent double-booking, validated with a dedicated concurrent race-condition test.
- Designed a secure payment workflow that confirms bookings only through verified Stripe webhooks rather than client-side state, closing a common payment/booking-mismatch risk.
- Built a multi-tenant authorization model using Supabase Row Level Security, covered by dedicated integration tests for database access policies.
- Developed a standalone Node.js/Express webhook service to handle Stripe events and Resend email notifications independently of the main application.
- Implemented instant-book and request-to-book workflows, availability calendars, image uploads, Stripe checkout, buyer/seller dashboards, earnings tracking, reviews, and cancellations.
- Built server-rendered pages with WCAG 2.1 AA accessibility considerations and dark/light theming.
- Wrote 500+ unit, component, and integration tests and maintained a zero-TypeScript-error, clean production build.

DocuQuery — Retrieval-Augmented Generation (RAG) Application (Independent Project)

Technologies: FastAPI, Python, Redis Streams, Chroma Cloud, MongoDB, PyMuPDF, OpenRouter, Docker, Render, React 19, React Router v8, TypeScript, Tailwind CSS v4, Vite

Links: github.com/bryankiansarmen/docuquery-api | github.com/bryankiansarmen/docuquery-web | docuquery-web.vercel.app

- Built a full-stack RAG application that allows users to upload PDF documents and ask natural-language questions grounded in their content.
- Developed an asynchronous FastAPI backend using Redis Streams and a dedicated worker process for document-processing workflows.
- Implemented hybrid search combining SPLADE sparse vectors and BGE-M3 dense vectors via Chroma Cloud, merged using Reciprocal Rank Fusion to improve retrieval relevance.
- Added Redis and MongoDB caching for exact and semantic query matches to reduce repeated language-model calls and retrieval latency.
- Designed a layered architecture separating API routes, services, and external clients for maintainability.
- Implemented per-key Redis rate limiting, SHA-256 duplicate-document detection, and persistent 10-turn conversation context using MongoDB.
- Deployed the API on Render using Docker with separate web-service and background-worker processes.
- Integrated OpenRouter chat and embedding models, with DeepSeek configured as the default language model.
- Built a React client with server-side rendering, dark mode, real-time upload-status polling, per-document sessions, and API connection monitoring.
- Maintained an enforced minimum 80% test coverage across 20+ unit, integration, worker, and end-to-end tests.

EDUCATION

Bachelor of Science in Information Systems

Immaculate Conception I-College of Arts and Technology, Sta. Maria, Bulacan, Philippines | Graduated June 2022

- Honors: Best in Capstone Project
- Recognition: Programmer of the Year